

About and queries

SQL Books Order Analysis

Project Overview

This project analyzes a **Book Sales** dataset using **Microsoft SQL Server (MS SQL)**. The dataset consists of three related tables: **Books**, **Customers**, and **Orders**. The objective of this project is to perform data cleaning, data exploration, and business analysis using SQL queries to extract meaningful insights from the data.

Database Creation and Data Import

A new database was created in **Microsoft SQL Server (MS SQL)**, and the following three CSV files were imported as separate tables:

1. Books Table

The **Books** table contains information about the books available for sale. It includes details such as:

- Book ID
- Title
- Author
- Genre
- Published Year
- Price
- Stock

Purpose:

This table stores the book inventory and is used to analyze book availability, pricing, genres, authors, publication years, and remaining stock after sales.

2. Customers Table

The **Customers** table stores information about customers who purchase books. It includes details such as:

- Customer ID
- Customer Name
- City
- Country

Purpose:

This table is used to analyze customer demographics, identify customers based on location, and evaluate customer purchasing behavior.

3. Orders Table

The **Orders** table records all book purchase transactions. It includes details such as:

- Order ID
- Book ID
- Customer ID
- Order Date
- Quantity
- Total Amount

Purpose:

This table links customers with the books they purchased and is used to analyze sales, revenue, order trends, and customer buying patterns.

Data Exploration and Data Cleaning

After importing the data, the following validation and cleaning steps were performed:

- Viewed all three tables using the **SELECT** statement.

Results		Messages					
	Book_ID	Title	Author	Genre	Published_Year	Price	Stock
1	1	Configurable modular throughput	Joseph Crane	Biography	1949	21.3400001525879	100
2	2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.7999992370605	19
3	3	Streamlined coherent initiative	Derrick Howard	Non-Fiction	1913	15.75	27
4	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.5200004577637	8
5	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.9499998092651	16
6	6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.55999994277954	2
7	7	Open-architected exuding structure	Jacqueline Young	Romance	1927	43.6300010681152	95
8	8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.9900016784668	84
9	9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.3299999237061	70
10	10	Ergonomic national hub	Samantha Ruiz	Mystery	2015	24.6299991607666	25
11	11	Secured zero tolerance time-frame	Denise Barnes	Fantasy	1998	35.9500007629395	10
12	12	Polarized optimal array	Destiny Scott	Non-Fiction	1989	27.4300003051758	63
13	13	Adaptive 5thgeneration orchestration	Jaclyn Miller	Romance	1913	14.039999961853	99
14	14	Re-engineered demand-driven parallelism	Jeremy Hayes	Science Fiction	1933	6.03999996185303	95
15	15	User-friendly motivating strategy	Keith Smith	Non-Fiction	1997	23.8299999237061	58
16	16	Vision-oriented tangible project	Christopher Price	Mystery	1941	10.0699996948242	8
17	17	Reduced secondary core	Benjamin Peters	Fantasy	1966	5.36999988555908	45
18	18	Adaptive 4thgeneration concept	Hector Palmer	Non-Fiction	2021	39.4700012207031	32

Results		Messages				
	Customer_ID	Name	Email	Phone	City	Country
1	1	Deborah Griffith	balljoseph@wright-keith.net	1234567891	South Craigfort	Denmark
2	2	Crystal Clements	kimberlybennett@curtis.com	1234567892	East Derekberg	Nicaragua
3	3	Susan Fuller	beanmichael@burnett-stewart.net	1234567893	Austinbury	Equatorial Guinea
4	4	Jamie Ramirez	amandahood@warren.com	1234567894	Dianamouth	Slovenia
5	5	Marcus Murphy	connerjohn@yahoo.com	1234567895	Smithbury	Guinea-Bissau
6	6	Stephen Vasquez	ricemiguel@yahoo.com	1234567896	Hamiltonstad	Rwanda
7	7	Susan Hicks	jeffrey91@yahoo.com	1234567897	East Rebecca	Montenegro
8	8	Matthew Johnson	austinkenneth@manning.net	1234567898	Kirstenborough	Israel
9	9	Matthew Williams	jeffrey41@diaz.com	1234567899	Rebeccafurt	Somalia
10	10	Ronald Osborn	staciekelley@heath.com	1234567900	Lake Benjamin	Cameroon
11	11	Thomas Garcia	rmiller@gmail.com	1234567901	West Monicabury	Argentina
12	12	Jennifer Murray	wilsonbrittany@hotmail.com	1234567902	South Ashleychester	Zimbabwe
13	13	Kristine Kim	sarahwilliams@hotmail.com	1234567903	Lake Robert	Nigeria
14	14	John Wood	johnsonalexander@gmail.com	1234567904	Richardsonville	New Caledonia
15	15	Vanessa Gaines	tbullock@gmail.com	1234567905	Rodriguezmouth	Wallis and Futuna
16	16	Stacey Flores	jackjackson@hotmail.com	1234567906	East Michaelfurt	Saint Helena
17	17	Christine Maldonado	ztaylor@yahoo.com	1234567907	Smithborough	Chile
18	18	David Watson	laurenlawson@alvarez-mcfarland.com	1234567908	New Josepbtown	Equatorial Guinea

Results		Messages				
	Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	1	84	169	2023-05-26	8	188.559997558594
2	2	137	301	2023-01-23	10	216.600006103516
3	3	216	261	2024-05-27	6	85.5
4	4	433	343	2023-11-25	7	301.209991455078
5	5	14	431	2023-07-26	7	136.360000610352
6	6	439	119	2024-10-11	5	249.399993896484
7	7	195	467	2023-10-23	6	82.9199981689453
8	8	32	159	2024-05-07	4	144.839996337891
9	9	109	407	2024-01-04	9	379.709991455078
10	10	94	122	2024-07-09	4	123
11	11	131	206	2023-10-16	1	38.0099983215332
12	12	454	3	2024-06-17	2	31.5
13	13	420	180	2023-06-08	5	125.449996948242
14	14	454	319	2023-08-24	2	85.2200012207031
15	15	127	479	2023-01-10	6	229.619995117188
16	16	412	196	2023-10-06	8	53.5200004577637
17	17	462	481	2023-03-20	5	52.75
18	18	377	101	2024-08-07	4	193.960006713867

Checked for duplicate records in:

- Books table
- Customers table
- Orders table

```

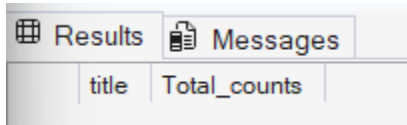
--checking duplicates in books table
select title, count(title) as Total_counts
from books
group by title
having count(title)>1;

--checking duplicates in Customers table

select Customer_ID, count(Customer_ID) as Total_counts
from Customers
group by Customer_ID
having count(Customer_ID)>1;

--checking duplicates in orders table
select Order_ID, count(Order_ID) as Total_Orders
from Orders
group by Order_ID
having count(Order_ID)>1;

```



Results	Messages
title	Total_counts

Checked for blank/null values in all three tables.

SQL Analysis

The following business questions were solved using SQL:

Found the total number of books in the **Fiction** genre.

```

--find fiction books
select * from books where genre='Fiction'

```

Results Messages							
	Book_ID	Title	Author	Genre	Published_Year	Price	Stock
1	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.5200004577637	8
2	22	Multi-layered optimizing migration	Wesley Escobar	Fiction	1908	39.2299995422363	78
3	28	Expanded analyzing portal	Lisa Coffey	Fiction	1941	37.5099983215332	79
4	29	Quality-focused multi-tasking challenge	Katrina Underwood	Fiction	1905	31.1200008392334	100
5	31	Implemented encompassing conglomeration	Melissa Taylor	Fiction	2010	21.2299995422363	44
6	39	Optimized national process improvement	Megan Goodwin	Fiction	1978	10.9899997711182	42
7	40	Adaptive didactic interface	Natalie Gonzalez	Fiction	1923	25.9699993133545	94
8	47	Reverse-engineered directional conglomer...	John Christian	Fiction	2006	20.3700008392334	90
9	62	Re-contextualized real-time strategy	Nicole Lynch	Fiction	1953	26.3400001525879	23
10	63	Polarized heuristic database	Franklin Mack	Fiction	1989	22.3799991607666	56
11	100	Synchronized client-server service-desk	James Alvarado	Fiction	1906	49.8899993896484	29
12	116	Multi-tiered foreground contingency	Jamie Gates	Fiction	1938	41.8199996948242	50
13	125	Public-key analyzing Graphic Interface	Abigail Madden	Fiction	1990	32.4099998474121	16
14	130	Realigned context-sensitive pricing structure	Jason Rodriguez	Fiction	2004	6.6399998664856	90
15	134	Polarized bandwidth-monitored throughput	Linda Newman	Fiction	1955	35.7200012207031	49

Retrieved all books published after **1950**.

--find books published after 1950

select * from books where Published_Year>1950

Results Messages							
	Book_ID	Title	Author	Genre	Published_Year	Price	Stock
1	2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.7999992370605	19
2	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.5200004577637	8
3	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.9499998092651	16
4	6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.55999994277954	2
5	8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.9900016784668	84
6	9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.3299999237061	70
7	10	Ergonomic national hub	Samantha Ruiz	Mystery	2015	24.6299991607666	25
8	11	Secured zero tolerance time-frame	Denise Barnes	Fantasy	1998	35.9500007629395	10
9	12	Polarized optimal array	Destiny Scott	Non-Fiction	1989	27.4300003051758	63
10	15	User-friendly motivating strategy	Keith Smith	Non-Fiction	1997	23.8299999237061	58
11	17	Reduced secondary core	Benjamin Peters	Fantasy	1966	5.36999988555908	45
12	18	Adaptive 4thgeneration concept	Hector Palmer	Non-Fiction	2021	39.4700012207031	32
13	19	Progressive asymmetric Internet solution	Sean Miller	Science Fiction	1990	11.3100004196167	1
14	20	Face-to-face systematic throughput	Teresa Brennan	Non-Fiction	1978	48.1300010681152	64
15	23	Reverse-engineered context-sensitive hardware	Christina Hernandez	Mystery	1967	38.5499992370605	70

Listed all customers from **Canada**.

--List all customers from canada

select * from Customers where country='Canada'

Results Messages						
	Customer_ID	Name	Email	Phone	City	Country
1	38	Nicholas Harris	christine93@perkins.com	1234567928	Davistown	Canada
2	415	James Ramirez	robert54@hall.com	1234568305	Maxwelltown	Canada
3	468	David Hart	stokesrebecca@gmail.com	1234568358	Thompsonfurt	Canada

Displayed all orders placed in **November 2023**.

```
--show orders placed in november 2023
select * from orders where order_date between '2023-10-30' and '2023-12-01'
```

Results		Messages				
	Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	4	433	343	2023-11-25	7	301.209991455078
2	19	496	60	2023-11-17	9	316.260009765625
3	75	291	375	2023-11-30	5	170.75
4	132	469	333	2023-11-22	7	194.320007324219
5	137	474	471	2023-11-25	8	363.040008544922
6	163	207	384	2023-11-23	3	101.76000213623
7	182	129	293	2023-11-01	7	125.51000213623
8	200	313	303	2023-11-23	1	6.57000017166138
9	213	325	447	2023-11-17	7	253.75
10	231	22	384	2023-11-11	1	33.9199981689453
11	245	386	97	2023-11-01	9	411.660003662109
12	252	405	387	2023-11-15	5	237.100006103516
13	257	123	403	2023-11-06	1	15.0100002288818
14	288	6	128	2023-11-13	1	24.0400009155273
15	307	368	133	2023-11-17	1	20.9599990844727
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Converted the **Price** column in the **Books** and **Orders** tables to **two decimal places**.

```
--converting price column in all 2 table till 2 decimal place
-- Permanently alter books
ALTER TABLE books
ALTER column price DECIMAL(10,2);
```

```
-- Permanently alter orders
ALTER TABLE Orders
ALTER COLUMN Total_Amount DECIMAL(10,2);
```

	Book_ID	Title	Author	Genre	Published_Year	Price	Stock
1	1	Configurable modular throughput	Joseph Crane	Biography	1949	21.34	100
2	2	Persevering reciprocal knowledge user	Mario Moore	Fantasy	1971	35.80	19
3	3	Streamlined coherent initiative	Derrick Howard	Non-Fiction	1913	15.75	27
4	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
5	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
6	6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
7	7	Open-architected exuding structure	Jacqueline Young	Romance	1927	43.63	95
8	8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84
9	9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.33	70
10	10	Ergonomic national hub	Samantha Ruiz	Mystery	2015	24.63	25
11	11	Secured zero tolerance time-frame	Denise Barnes	Fantasy	1998	35.95	10
12	12	Polarized optimal array	Destiny Scott	Non-Fiction	1989	27.43	63
13	13	Adaptive 5thgeneration orchestration	Jaclyn Miller	Romance	1913	14.04	99
14	14	Re-engineered demand-driven parallelism	Jeremy Hayes	Science Fiction	1933	6.04	95
15	15	User-friendly motivating strategy	Keith Smith	Non-Fiction	1997	23.83	58

Calculated the total stock of books available.

```
--find total stock of books available
select sum(stock) as total_stock from Books
```

	Book_ID	Title	Author	Genre	Published_Year	Price	Stock
1	1	Configurable n	Joseph Crane	Biography	1949	21.34	100
2	2	Persevering re	Mario Moore	Fantasy	1971	35.80	19
3	3	Streamlined co	Derrick Howard	Non-Fiction	1913	15.75	27
4	4	Customizable 24hour product	Christopher Andrews	Fiction	2020	43.52	8
5	5	Adaptive 5thgeneration encoding	Juan Miller	Fantasy	1956	10.95	16
6	6	Advanced encompassing implementation	Bryan Morgan	Biography	1985	6.56	2
7	7	Open-architected exuding structure	Jacqueline Young	Romance	1927	43.63	95
8	8	Persistent local encoding	Troy Cox	Science Fiction	2019	48.99	84
9	9	Optimized interactive challenge	Colin Buckley	Fantasy	1987	14.33	70
10	10	Ergonomic national hub	Samantha Ruiz	Mystery	2015	24.63	25
11	11	Secured zero tolerance time-frame	Denise Barnes	Fantasy	1998	35.95	10
12	12	Polarized optimal array	Destiny Scott	Non-Fiction	1989	27.43	63
13	13	Adaptive 5thgeneration orchestration	Jaclyn Miller	Romance	1913	14.04	99
14	14	Re-engineered demand-driven parallelism	Jeremy Hayes	Science Fiction	1933	6.04	95
15	15	User-friendly motivating strategy	Keith Smith	Non-Fiction	1997	23.83	58

Retrieved the details of the most expensive book.

```
--Detail of most expensive book
select top 1 * from Books order by price desc
```

Results		Messages					
	Book_ID	Title	Author	Genre	Published_Year	Price	Stock
1	340	Proactive system-worthy orchestration	Robert Scott	Mystery	1907	49.98	88

Listed all customers who ordered more than one quantity.

```
--All customers who ordered more than 1 quantity of book
select * from Orders where Quantity>1
```

Results		Messages				
	Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21
5	5	14	431	2023-07-26	7	136.36
6	6	439	119	2024-10-11	5	249.40
7	7	195	467	2023-10-23	6	82.92
8	8	32	159	2024-05-07	4	144.84
9	9	109	407	2024-01-04	9	379.71
10	10	94	122	2024-07-09	4	123.00
11	12	454	3	2024-06-17	2	31.50
12	13	420	180	2023-06-08	5	125.45
13	14	454	319	2023-08-24	2	85.22
14	15	127	479	2023-01-10	6	229.62
15	16	412	196	2023-10-06	8	53.52

Displayed all orders where the total amount was greater than \$20.

```
--all orders where total_amount is more than 20
select * from orders where Total_amount>20
```

Results

Messages

	Order_ID	Customer_ID	Book_ID	Order_Date	Quantity	Total_Amount
1	1	84	169	2023-05-26	8	188.56
2	2	137	301	2023-01-23	10	216.60
3	3	216	261	2024-05-27	6	85.50
4	4	433	343	2023-11-25	7	301.21
5	5	14	431	2023-07-26	7	136.36
6	6	439	119	2024-10-11	5	249.40
7	7	195	467	2023-10-23	6	82.92
8	8	32	159	2024-05-07	4	144.84
9	9	109	407	2024-01-04	9	379.71
10	10	94	122	2024-07-09	4	123.00
11	11	131	206	2023-10-16	1	38.01
12	12	454	3	2024-06-17	2	31.50
13	13	420	180	2023-06-08	5	125.45
14	14	454	319	2023-08-24	2	85.22
15	15	127	479	2023-01-10	6	229.62
16	16	412	196	2023-10-06	8	53.52
17	17	462	481	2023-02-20	5	52.75

Listed all genres available in the **Books** table.

```
--list all genre available in books table
select distinct(genre) from Books
```

Results		Messages	
	genre		
1	Biography		
2	Fantasy		
3	Fiction		
4	Mystery		
5	Non-Fiction		
6	Romance		
7	Science Fiction		

Retrieved the books with the lowest stock.

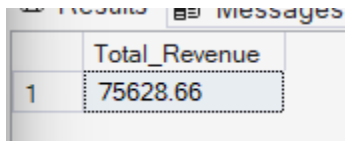
```
--book with lowest stock
select top 1 * from books order by Stock ASC
```

	Book_ID	Title	Author	Genre	Published_Year	Price	Stock
1	44	Networked systemic implementation	Ryan Frank	Science Fiction	1965	13.55	0

Calculated the total revenue generated from all orders.

```
--TOTAL REVENUE GENERATED FROM ALL ORDERS
```

```
select sum(total_amount) as Total_Revenue from Orders
```

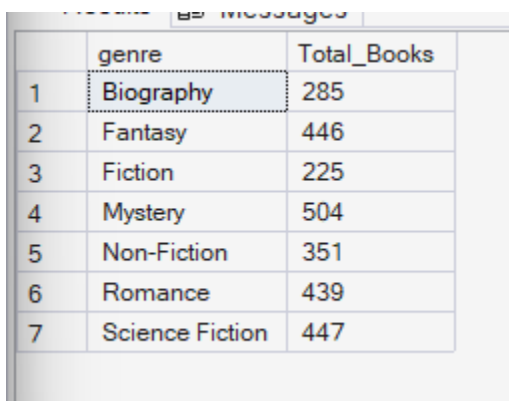


	Total_Revenue
1	75628.66

Calculated the total number of books sold for each genre.

```
--Total Number of books sold for each genre
```

```
select b.genre, sum(o.quantity) as Total_Books  
from books b  
join orders o  
on b.Book_ID=o.Book_ID  
group by genre
```



	genre	Total_Books
1	Biography	285
2	Fantasy	446
3	Fiction	225
4	Mystery	504
5	Non-Fiction	351
6	Romance	439
7	Science Fiction	447

Calculated the average price of books in the **Fantasy** genre.

```
--avg price for genre='fantasy'
```

```
select genre, AVG(price)  
from Books  
where genre='fantasy'  
group by genre
```

	genre	(No column name)
1	Fantasy	25.981690

Listed customers who placed at least two orders along with their total number of orders.

```
select c.customer_id, c.name, count(o.order_id) as Total_orderss
from Customers c
join Orders o
on c.Customer_ID=o.Customer_ID
group by c.Customer_ID, c.Name
having count(o.order_id)>1
order by Total_orderss asc
```

	customer_id	name	Total_orderss
1	214	Alexander Scott	3
2	119	Alyssa Cuevas	3
3	239	Amanda Kramer	2
4	107	Amy Hunt	4
5	59	Andrew Collins	3
6	485	Andrew Figueroa	4
7	420	Andrew Murray	3
8	467	Anna Green	2
9	284	Annette Garcia	2
10	378	Anthony Kane	2
11	199	Anthony Simmons	2
12	474	Anthony Young	5
13	454	April Anderson	2
14	103	Ashley Castro	2
15	121	Ashley Hernandez	2
16	425	Ashley Perez	4
17	462	Brendan Dunn	2

Identified the most frequently ordered book.

```
select top 7 o.book_id, b.Title, count(o.Order_ID) as Total_order_Times
from Orders o
join Books b
on b.Book_ID= o.Book_ID
group by o.book_id, b.Title
order by Total_order_Times desc
```

Results

Messages

	book_id	Title	Total_order_Times
1	333	Advanced responsive extranet	4
2	273	Devolved zero administration process improvement	4
3	31	Implemented encompassing conglomeration	4
4	120	Integrated secondary access	4
5	491	Pre-emptive intangible adapter	4
6	73	Realigned multi-tasking installation	4
7	88	Robust tangible hardware	4

Retrieved the top three most expensive books in the **Fantasy** genre.

```
--Top 3 most expensive books of fantasy genre
select top 3 title, price
from books
where genre='fantasy'
order by price desc
```

	title	price
1	Stand-alone content-based hub	49.90
2	Innovative 3rdgeneration database	49.23
3	Optimized even-keeled analyzer	48.97

Calculated the total quantity of books sold by each author.

```
--Total quantity of books sold by each author
select * from orders
select b.author , sum(o.Quantity) as total_books
from books b
join orders o
on o.book_id =b.book_id
group by b.author
order by total_books desc
```

Results		Messages
	author	total_books
1	Patrick Contreras	28
2	Melissa Taylor	27
3	Emily James	24
4	Thomas Trujillo	24
5	Valerie Moore	23
6	Ellen Doyle	23
7	Erica Parker	23
8	Sheena Harris	23
9	Rachel Gibbs	22
10	Amanda Wilson	22
11	Anna Roberts	21
12	Kristi Phillips	21
13	Tonya Saunders	21
14	Michael Hill	20
15	Alexander Wall...	20
16	Joyce Patton	19
17	Paul Miller	18

Listed the cities where customers spent more than \$30.

```
--Cities where customers are located who spent more than $30
```

```
select o.Customer_ID, c.Name, c.city as City_name, sum(o.total_amount) as Total_spending
from orders o
join Customers c
on o.Customer_ID=c.Customer_ID
group by o.Customer_ID, c.Name, c.city
having sum(o.total_amount)>30
order by Total_spending asc
```

	Customer_ID	Name	City_name	Total_spendings
1	203	William Walters	South Heather	30.02
2	309	Scott Lee	East Tracystad	30.02
3	39	Tonya Arnold	Nelsonmouth	31.48
4	310	Jennifer Lopez	North Emily	31.68
5	447	Emily Wilkerson	Parkerside	32.82
6	44	Alexis Gallegos	Conniefort	33.45
7	261	David Donovan	West Justin	34.16
8	360	Roberto Martinez	Patriciaville	35.49
9	12	Jennifer Murray	South Ashleychester	39.51
10	31	Amanda Morton	Marktown	41.80
11	206	Diana Davis	Christophertown	42.10
12	161	Mr. Frederick Harrison	Darlenehaven	42.19
13	294	Wesley Parks	Angelaside	42.19
14	333	Valerie Miller	North Rodney	42.58
15	344	Shawn Allen Jr.	Port Bruce	43.12
16	247	Michael Ellis	Port Austinview	43.12

Retrieved the top 10 customers who spent the most on orders.

```
-- Top 10 Customers who spend most on orders
select top 10 c.name, c.customer_id, sum(o.total_amount) as Total_spendings
from customers c
join orders o
on c.Customer_ID =o.Customer_ID
group by c.name, c.customer_id
order by Total_spendings desc
```

	name	customer_id	Total_spendings
1	Kim Turner	457	1398.90
2	Jonathon Strickland	174	1080.95
3	Carrie Perez	364	1052.27
4	Julie Smith	405	991.00
5	Pamela Gordon	386	986.30
6	Ashley Perez	425	942.62
7	Anthony Young	474	929.19
8	Robert Clark	163	746.65
9	Justin Spencer	167	719.93
10	Alexander Scott	214	682.15

Calculated the remaining stock after all orders were completed.

```
--calculate stock remaining after all orders are done
select b.book_id , b.Title, b.Stock,sum(o.quantity) as Total_Books_Ordered,
coalesce(b.Stock-sum(o.quantity),0) as remaining_books
from books b
join orders o|
on b.Book_ID=o.Book_ID
group by b.book_id, b.Title, b.Stock
having coalesce(b.Stock-sum(o.quantity),0)>0
```

110 % No issues found

Results

Messages

	book_id	Title	Stock	Total_Books_Ordered	remaining_books
1	1	Configurable modular throughput	100	3	97
2	3	Streamlined coherent initiative	27	5	22
3	5	Adaptive 5thgeneration encoding	16	8	8
4	7	Open-architected exuding structure	95	5	90
5	8	Persistent local encoding	84	3	81
6	10	Ergonomic national hub	25	1	24
7	11	Secured zero tolerance time-frame	10	5	5
8	13	Adaptive 5thgeneration orchestration	99	9	90
9	16	Vision-oriented tangible project	8	1	7
10	17	Reduced secondary core	45	9	36
11	18	Adaptive 4thgeneration concept	32	14	18
12	21	Down-sized static interface	87	3	84
13	25	Devolved mobile conglomeration	79	6	73
14	26	Multi-channeled multi-tasking capability	41	11	30
15	28	Expanded analyzing portal	79	2	77
16	29	Quality-focused multi-tasking challenge	100	4	96

Code

--Viewing all 3 tables

```
select * from Books
```

```
select * from Customers
```

```
Select * from Orders
```

--checking duplicates in books table

```
select title, count(title) as Total_counts
```

```
from books
```

```
group by title
```

```
having count(title)>1;
```

--checking duplicates in Customers table

```
select Customer_ID, count(Customer_ID) as Total_counts
```

```
from Customers
```

```
group by Customer_ID
```

```
having count(Customer_ID)>1;
```

--checking duplicates in orders table

```
select Order_ID, count(Order_ID) as Total_Orders
```

```
from Orders
```

```
group by Order_ID
```

```
having count(Order_ID)>1;
```

--checking blanks in books table

```
select Book_ID
```

```
from Books
```

```
where Book_ID=""
```

--checking blanks in customer table

```
select Customer_ID
```

```
from Customers
```

```
where Customer_ID=""
```

--checking blanks in orders table

```
select Order_ID
```

```
from Orders
```

```
where Order_ID=""
```

--find fiction books

```
select * from books where genre='Fiction'
```

--find books published after 1950

```
select * from books where Published_Year>1950
```

--List all customers from canada

```
select * from Customers where country='Canada'
```

--show orders placed in november 2023

```
select * from orders where order_date between '2023-10-30' and '2023-12-01'
```

--converting price column in all 2 table till 2 decimal place

```

-- Permanently alter books
ALTER TABLE books
ALTER column price DECIMAL(10,2);

-- Permanently alter orders
ALTER TABLE Orders
ALTER COLUMN Total_Amount DECIMAL(10,2);

--find total stock of books available
select sum(stock) as total_stock from Books

--Detail of most expensive book
select top 1 * from Books order by price desc

--All customers who ordered more than 1 quantity of book
select * from Orders where Quantity>1

--all orders where total_amount is more than 20
select * from orders where Total_amount>20

--list all genre available in books table
select distinct(genre) from Books

--book with lowest stock
select top 1 * from books order by Stock ASC

--TOTAL REVENUE GENERATED FROM ALL ORDERS
select sum(total_amount) as Total_Revenue from Orders

--Total Number of books sold for each genre
select b.genre, sum(o.quantity) as Total_Books
from books b
join orders o
on b.Book_ID=o.Book_ID
group by genre

--avg price for genre='fantasy'
select genre, AVG(price)
from Books
where genre='fantasy'
group by genre

select * from orders
--list of people who placed at least 2 orders along with their total orders
select c.customer_id, c.name, count(o.order_id) as Total_orderss
from Customers c
join Orders o
on c.Customer_ID=o.Customer_ID
group by c.Customer_ID, c.Name
having count(o.order_id)>1
order by Total_orderss asc

```

--most frequently ordered book

```
select top 7 o.book_id, b.Title, count(o.Order_ID) as Total_order_Times
from Orders o
join Books b
on b.Book_ID= o.Book_ID
group by o.book_id, b.Title
order by Total_order_Times desc
```

--Top 3 most expensive books of fantasy genre

```
select top 3 title, price
from books
where genre='fantasy'
order by price desc
```

--Total quantity of books sold by each author

```
select * from orders
select b.author , sum(o.Quantity) as total_books
from books b
join orders o
```

```
on o.book_id =b.book_id
group by b.author
order by total_books desc
```

--Cities where customers are located who spent more than \$30

```
select o.Customer_ID, c.Name, c.city as City_name, sum(o.total_amount) as Total_spending
from orders o
join Customers c
on o.Customer_ID=c.Customer_ID
group by o.Customer_ID, c.Name, c.city
having sum(o.total_amount)>30
order by Total_spending asc
```

-- Top 10 Customers who spend most on orders

```
select top 10 c.name, c.customer_id, sum(o.total_amount) as Total_spending
from customers c
join orders o
on c.Customer_ID =o.Customer_ID
group by c.name, c.customer_id
order by Total_spending desc
```

--calculate stock remaining after all orders are done

```
select b.book_id , b.Title, b.Stock,sum(o.quantity) as Total_Books_Ordered,
coalesce(b.Stock-sum(o.quantity),0) as remaining_books
from books b
join orders o
on b.Book_ID=o.Book_ID
group by b.book_id, b.Title, b.Stock
having coalesce(b.Stock-sum(o.quantity),0)>0
```

